

Schrödinger’s Update: User Perceptions of Uncertainties in Proprietary Large Language Model Updates

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ABSTRACT

Developers of proprietary large language models (LLMs), like OpenAI and Anthropic, often roll out updates, some announced and others silent, impacting transparency for users. We interviewed 21 LLM users, including end-users, developers, and academics, mainly using ChatGPT and GPT4. Many reported feelings of loss of control and uncertainty, and diminished trust due to silent updates and deprecations. Participants overall felt that proprietary LLMs were similar to traditional software with regards to updates, only differing in the impossibility in full transparency and explainability. Consequently, participants felt LLM companies departed from established software update norms without justification, in some cases appealing to the novelty of AI in order to excuse bad practices. As a result, users are inclined to shift towards fixed, open-source models, reserving proprietary LLMs for prototyping. We suggest strategies that LLM providers can adopt to better uphold transparency and trust during model updates and deprecations.

CCS CONCEPTS

• Human-centered computing → User studies.

KEYWORDS

Large Language Models, Software Updates, Proprietary Models

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1 INTRODUCTION

“ChatGPT used to provide nicely markdown formatted code. But after an update a month ago, it stopped producing code in that markdown block. This issue began just 2 weeks before I had to test with actual users. That was a very, very stressful moment.”

–Participant P11-a

Text generation large language models (LLMs) represent major advancements in natural language interaction [7, 8]. At the forefront of these advances are companies like OpenAI and Anthropic, which offer “proprietary” LLMs through chat interfaces and programmatic APIs: powerful, proprietary models like ChatGPT that end-users and developers alike may use for cross-domain tasks. Yet the secretive, opaque practices of proprietary LLM companies can cause significant frustration. The disruptive experience of *P11-a* above, an academic researcher that we interviewed, is only one instance. Chen et al. [12] observed silent degradation and undocumented performance changes in OpenAI models; similarly, Anthropic’s release of Claude-2 coincided with the removal of specific model names from their API, rendering previous models inaccessible.¹ From past work, it is known that unclear software updates erode user trust [18, 64], disrupts dependent software systems [15], and causes distress [36]; however, such work has not yet covered LLMs in particular.

In this article, we seek to understand how end-users, software developers, and academics perceive the model update practices of proprietary LLM companies. What makes LLM updates distinctly *different* from traditional software updates? Are they? If so, to what extent? We interviewed 21 participants, ranging from end-users to developers and researchers who primarily used ChatGPT and GPT4. Our research shows that silent updates and removals in LLM APIs reduce trust among academics and developers. This leads to individually unique yet time-consuming practices for testing prompts. Additionally, concerns about study replicability and software stability cause a shift: users now use proprietary LLMs only for rapid

¹Though they were still accessible if one knew the names, it was also unclear for how long they will remain supported. This anecdote is from first-hand experience of the last author, who is the lead developer of a prompt engineering tool and had to integrate Anthropic’s changes.

prototyping but seek fixed, open-source models for research and development.

Although we initially hypothesized that participants would emphasize major differences between LLMs and traditional software given the revolutionary rhetoric about AI [8], we found that participants mainly reflected on their similarities while voicing concerns for nontransparency. Overall, participants believed that proprietary LLM companies like OpenAI violated well-established software update practices without justification. They are concerned that *AI companies can use the novelty surrounding AI to excuse bad practices*. Building the comparison between subjective perceptions and existing literature on software updates, we embark on a discussion of how proprietary LLMs appear to be very similar to traditional software with regards to updates, yet lacking clarification and transparency.

Consequently, we raise concerns that the “move fast and break things” culture of companies like OpenAI may, if left unchecked, can cause a mass exodus from proprietary LLM providers when and if models that users have total control over become fast, performant, and cheap enough to integrate instead (e.g., open-source, local models). To rectify this, companies should adopt and learn from standard software updating practices, only deviating for the aspects of LLMs that are actually different from prior software (e.g., impossibility of full explainability).

2 RELATED WORK

LLMs are language models that can generate human-like text, answer questions, assist in writing [22], help with coding [48], translate languages [28], and tutor in various areas [11, 47]. While some touted their capabilities as the emergence of Artificial General Intelligence [8], they are still software: they are iteratively developed, deployed, and updated. We thus first provide an overview of the extensive literature on software updates, before leading into recent developments in LLMs and how their unique properties might differ from prior types of software updates.

2.1 Software Updates

Software updates, which introduce changes to programs by addressing issues, adding features, or modifying functionality, are often approached by users cautiously. Users move through six stages: awareness, deciding to install, preparation, installation, troubleshoot, and post-state [64]. In early stages, like “awareness” and “deciding to install,” user reluctance stems from distrust of software companies or fears of malware [3, 52]. This hesitation persists, with users extensively preparing for potential issues in later stages, such as data loss [33], or being wary of new features that change existing functionality [16, 46].

Not just end-users, but developers too are hesitant to adopt updates, especially with APIs. For example, despite Android’s rapid API update with an average of 115 updates per month, there’s a lag in developer adoption, with about 28% of API references in client applications being outdated and a median adaptation delay of 16 months. Furthermore, only 22% of these outdated API usages are updated, taking an average of 14 months, which is much longer than the typical 3-month API release interval [39]. Such lags contribute

to challenges for developers, as 80% of code changes disrupting client-side code are due to API refactoring [15, 54].

Failing to update software in a timely manner often exposes users to significant security risks such as heightened vulnerability to attacks, with unpatched issues potentially leading to system compromises, as regular updates minimize these vulnerabilities and bolster system defense [6, 30, 37, 51, 65]. To mitigate such risks and encourage users to update, developers have increasingly adopted automatic updating practices [24, 26, 43, 44, 68]. While convenient, these updates can compromise user agency and trust by sidelining user decisions [53, 73]. For instance, they might not always be compatible with accessibility features, posing challenges for users with disabilities [71]. Recognizing these challenges, researchers like Vaniea and Rashidi [64] advocate for more user-centric approaches, suggesting that users be kept informed about updates and provided with restore points for any undesired changes.

2.2 Updating Proprietary LLMs

Major LLM providers like OpenAI, Google, Anthropic, Microsoft, and Meta are in a competitive race to enhance their models, driven by market demands, regulatory concerns and the need for “AI safety”. They frequently update their LLMs to address safety risks, counteract malicious use, and improve efficiency [2, 5, 21, 27, 32, 35, 58, 59, 66, 67]. Yet, these updates are often lack transparency. From a technical angle, LLM developers are not able to document every granular changes due to LLMs’ deep neural network structures with billions of interacting parameters [7, 50]. Unlike deterministic code changes, LLM updates require processes such as fine-tuning [4, 25], where additional data are required to be incorporated to adjust the model’s behavior, refining the model’s responses and understanding. Even minor changes to the input data or training procedure can result in unpredictable and significant impacts on the model’s outputs [14, 29, 31].

Several anecdotal accounts and empirical studies have voiced transparency issues and frustrations deriving from LLM updates, especially around OpenAI [45]. Instances include the sudden discontinuation of Codex [13, 48], emotional distress from unexpected changes in apps like Replika [36], unannounced GPT-4 updates affecting performance [12], and Anthropic’s removal of previous model references with Claude-2’s introduction [1]. These practices deviate from recommended software maintenance protocols, which advocate for user engagement throughout the update process, as outlined by Vaniea & Rashidi [64], which identified six stages of software updates: awareness, decision-making, preparation, installation, troubleshooting, and post-installation, emphasizing user empowerment at each phase. However, LLM updates, particularly in proprietary models, tend to bypass the initial four stages, deploying or deprecating features without user consent or notification. Not all generative model providers limit user agency, however. MidJourney preserves its earlier image models for user flexibility [41], while Stability.AI and Meta provide open-source access to their image and language models [55, 62]. Here, we mainly focus on text generation models given their popularity and broad application. how do users and developers view and react to unpredictable LLM API changes? How does the transparency in LLM updates compare

to conventional software changes, and do users see these differences as warranted? How do hidden updates impact trust, and how do users manage these alterations? What do users believe are the most ethical ways to update proprietary LLMs, and how can these views inform future best practices?

3 METHODS

3.1 Recruitment

To study user perceptions of LLM updates, we disseminated a recruitment survey on HackerNews, OpenAI’s developer forum, and X. The survey collected demographics information and inquired about experiences with model updates, requesting GitHub or project portfolio links. Of 154 responses, we selected 10 ChatGPT users and 30 others with GitHub or portfolio links containing projects related to using LLM APIs. All invited users indicated that they were affected by updates. From the 40 invited for interviews, 21 participated, and notably, none declined directly.

3.2 Semi-structured interview

We conducted semi-structured interviews with 21 participants: 4 end users of ChatGPT, 9 developers, and 8 academic researchers using proprietary LLM APIs. The group included 10 males, 10 females, and 1 non-binary individual, with an average age of 28. Participants are denoted as “-e” for end users, “-d” for developers, and “-a” for academic researchers in our results (e.g., P11-a for an academic researcher).

Participants gave informed consent for interviews, with the option to withdraw, and were compensated with a \$30 Amazon Gift Card for their 30-45 minute participation. The focus was on their experiences with LLM updates, including disruptions and suggestions for improvement. Transcripts were anonymized, and the study was approved by our university’s Institutional Review Board; detailed interview guides are in the appendix A.

3.3 Data analysis

The two lead authors independently coded 2 interview transcripts from each of the ChatGPT user, developers and academic researchers using open coding [10]. This approach identified participants’ frustrations with model uncertainties, their coping strategies, and suggestions for LLM companies. Subsequently, the team created a codebook for analysis. The two lead authors then reviewed the remaining transcripts, refining the codebook until data saturation was reached. The complete codebook can be found in the appendix B

4 RESULTS

We organize our findings from current problems, to consequences, to future solutions: how users perceived uncertainties in model update practices, how these uncertainties changed their behavior (consequences), and user recommendations for how to improve updating practices in the future.

4.1 Uncertainties in Model Updates

4.1.1 Perceived model degradation without announcement. Participants expressed frustration over the perceived reduction in model performance, which they attributed to the growing user base and

server pressures. For instance, P14-a, an early user of GPT-4 API, observed a decline in GPT-4’s performance on certain standardized tasks: *“It used to be I could describe a web app, and it’d create a basic version. Now, I have to guide it like I’m working with a student to get what I want”* (P14-a). Similarly, P1-e also noticed the degradation in maneuverability in asking ChatGPT to generate the desired responses: *“I need to be a little bit more explicit now with the instructions that I give it in order to get the same types of responses.”*

4.1.2 Model optimization with unanticipated side effects. Participants found that while OpenAI might be optimizing certain model attributes, such as alignment, these optimizations inadvertently altered unique model behaviors they relied on. P11-a shared an instance where a model update stopped generating code in a specific markdown format, stating, *“ChatGPT used to provide nicely markdown formatted code. But after an update a month ago, it stopped producing code in that markdown block. This issue began just 2 weeks before I had to test with actual users. That was a very, very stressful moment”* (P11-a).

4.1.3 Abrupt model deprecation. Participants were troubled by the sudden loss of API access and the need to switch to alternative APIs, which often had functional differences. P13-a was particularly distressed by the short notice of Codex model’s deprecation: *“The week before our formal user studies started, we got this email ‘yeah, we’re deprecating Codex. You’re not gonna be able to use it anymore.’ So we were kind of in a panic.”* (P13-a). Participants compared their experiences to conventional software updates, emphasizing that traditional updates offer more advanced notice before deprecation compared to proprietary LLM APIs. *“You can specify Gpt-3.5-turbo version XXX, which is kind of a freeze of a point in time, but then after some time they will discontinue that checkpoint. You can’t really rely on anything for the long term.”* (P18-d).

4.1.4 Insufficient documentation for model changes. Participants were dissatisfied with the rapid model updates that lacked adequate documentation, leaving them without essential guidance for transitioning between versions. P13-a recounted their struggle with an OpenAI model update: *“[OpenAI] said, ‘Oh, it is no problem. You can just drop in 3.5-turbo and everything will just work.’ It didn’t. The GPT-3.5-turbo model would just not give us any completions at all. It was a mess”* (P13-a). P11-a agreed with P13-a, comparing OpenAI’s documentation to the informative blog posts found on online forums. *“OpenAI generally releases [model updates] as a blog post. But none of it actually helped me. Blog posts in Hacker News, talking about what other people have noticed in terms of changes to the model is a lot more useful to me because they generally talk about how to migrate models using prompt engineering”* (P11-a).

Compared to industry standards, participants found OpenAI’s documentation inadequate, leaving them to interpret updates without guidance. P19-d, a technological consultant for a large firm, emphasized the fast iterations of OpenAI’s models. This combination of minimal communication and fast-paced updates makes it challenging for his clients to adapt to the technology. *“[OpenAI models have] changed so fast, and they publish very little guidance about how to use these things. [...] I mean, I’ve worked with big vendors. I would say that [OpenAI’s] documentation is nonexistent compared to*

industry standards. [...] It's pretty much up to us, the users to go poke [the API] and see, 'does it handle this prompt right or not?'"

4.2 Consequences of uncertainties in model updates

4.2.1 Users' Developing Personalized Solutions by Themselves. The absence of clear guidelines from OpenAI about model updates forces users and developers to devise unique, personalized solutions to maintain consistent model outputs. However, these individualized fixes are not always effective, causing users significant time and efforts. For instance, P11-a detailed three methods they used to tackle disruptions caused by model updates: they either "added instructions to handle error cases", "recreated the whole prompt", or "handled errors outside of the LLM API call" if the other steps didn't work (P11-a). However, sometimes adjusting prompts isn't enough. For example, P9-a had to spend time on the tedious task of re-tuning hyper-parameters using a grid search after each update, instead of just re-optimizing prompts. This highlights the amount of time and effort people have to spend on fixing disruptions individually due to the lack of structured guidance.

4.2.2 Lost trust in proprietary LLM APIs. End-users, academic researchers, and developers had concerns over the ambiguity of OpenAI's model updates and communications. These concerns revolve around perceived performance degradations in GPT-4, despite claims of improvement, leading to a loss of trust and transparency issues. For academia researchers, the unpredictability of model updates poses risks to the reproducibility of research, as models may change or become inaccessible before studies are completed: "And now essentially, my whole research is hanging on one API call thread. Which can be cut off anytime. That is very bad." (P11-a) Developers feared that the long-term sustainability of applications built on these models due to potential changes in API interactions. These issues highlight a broader apprehension about the reliability and clarity of OpenAI's updates and their implications for various stakeholders: "You'd never know what you can expect ... If the new model comes along then you have to interact in a different way. I think this is the biggest disadvantage [of using proprietary LLM APIs] right now." (P18-d)

The ambiguity surrounding model updates, along with OpenAI's ambiguous communications regarding the effects and intentions behind them, left a spectrum of participants—including end-users, academic researchers, and developers—apprehensive about engaging with these models, leading to diminished trust and perceptions of reduced transparency. P2-e noted the inconsistency of GPT-4's perceived performance degradation in some domains and OpenAI's characterization of it as an improvement. "[OpenAI] made guardrails, improvements to GPT-4, which caused, in some domains, clear performance degradation, and then, claimed that ChatGPT is smarter. And that I think was just losing the public's trust. There's so many different notions of smarter. When he says smarter he's probably talking about behaving according to the guardrails'. But when people say smarter, they mean it's able to complete tasks." (P2-e).

4.2.3 Exploration of alternatives. Driven by the unpredictable nature of proprietary LLMs and their updates, some participants are

now considering only utilizing the APIs for prototyping or transitioning to open-source or local models (P18-a). P14-a echoed: "OpenAI is so closed. And I've always just seen it as a tool for prototyping, but not a tool that I will be ever comfortable with being a dependency in one of my systems. Because it's unreliable." (P14-a)

4.3 User-Recommended Improvements for LLM Updating Practices

4.3.1 Improve transparency of benchmarking, training data, and update changes. Some participants perceived increased complexity associated with the LLMs compared to traditional software, perceiving the key difference as the impossibility of full transparency. They perceived that traditional practices aimed at enhancing software transparency may inadequately address the challenges posed by LLM updates, owing to the complexity of LLMs and the diverse ways that users may interact with models. P9-a argued that traditional APIs are designed to complete "specific tasks". Therefore evaluating the functionalities or reliabilities of these APIs are easy. However, there's no "comprehensive benchmark to evaluate the LLMs, which are for wide range of different domains or tasks". Participants' varied contexts and practices led to differing standards for selecting a model based on key attributes, further complicating the task of choosing aspects to evaluate in a benchmark. For example, P13-a discussed criteria for assessing a model's suitability for code generation tasks: "What's important of a model is very qualitative and very subjective. For some, the fact that the code passing [the test] is obviously important, for others it might not matter very much." (P13-a)

To partially address these challenges, participants emphasized the importance of benchmarking models. For example, P19-d highlighted the need for universal benchmarks, stating that there should be standards "that every language model had to be scrutinized against." Building such benchmarks might require collaboration between LLM companies and third-party entities, such as U.S.'s Federal Communications Commission. Calls for increased transparency in model development were also prevalent. Participants, like P9-a, wanted more information on training processes and datasets, to better gauge a model's ability in specific domains. P9-a mentioned, "to check if a model has sufficient math capability, I would check if it was trained on pertinent math data." Furthermore, P12-a suggested ways for companies like OpenAI to enhance transparency during updates, advising that they provide guidance on model behavioral changes: "Guide us on the changes that should be made and how the codes should look like."

4.3.2 Behave like software companies and avoid excusing bad practices with overtures to novelty of AI. Participants perceived that LLM companies, especially OpenAI, believed themselves to be above or separate from traditional software companies, citing revolutionary rhetoric and hubris. They argued that companies need to begin acting like conventional software companies rather than justifying bad practices by overtures to the novelty of AI. For instance, P19-d expressed concerns about OpenAI's double roles, portraying itself as a research company while operating as a software vendor. "OpenAI has bit of hubris about their ability to achieve AGI, where they're literally just making one model. My advice is to focus on being a good vendor. [...] Their business objectives and their moral objectives and

their scientific objectives are just all out of alignment” (P19-d). This sentiment was further reinforced by the observation that although many people in the field consider LLMs as revolutionary, these companies’ core principles are fundamentally software-based. *“They think of [LLM] as new kind of software, but it’s still just software and companies are always judged by the quality of their documentation, the quality of their product. They think that they’re different. And of course, everyone thinks that they’re different when they’re the new kid on the block”* (P19-d).

4.3.3 Adopt conventional software company updating practices. Participants found incorporating traditional software engineering practices, including proper notification about updates and deprecation, systematic versioning of the models, documenting the updates, and onboard users important.

Participants argued that there needs to be full transparency in updates, and such norms should be set early. *“Just at the very first step would be full transparency on what the updates were and what the potential impacts of this update is. And I don’t see that currently.”* (P7-d).

Participants mentioned having a proper warning before any update: *“never, never change the model that an Api uses.”* (P14-a). Drawing parallels with mobile phone updates, P5-e commented, *“whenever your phone updates behind your back, it sends you a little blurb about what changed, right? That’s like transparency.”* Participants shared the sentiments, indicating that To elucidate this, P13-a shared, *“[Updating] is not just switching the name of a model in our code. We also need to test it out. For getting the same responses... [Prompt engineering] takes time, it takes many rounds of iteration and testing.”*

Participants mentioned the benefits of proper versioning, which allows them to revert. Reverting to older LLM versions is crucial due to model-sensitive prompts and the varying performance across versions, which may make an older model more desirable: *“LLMs will have a long way to catch up in terms of robustness and accuracy. It’s essential for me to have control over the versions because I’ve optimized [the prompts] after significant effort.”*

5 DISCUSSION

5.1 No Excuse to depart from conventional software update practices

Our results challenge the idea that LLMs should be exempt from standard software update conventions. Participants noted that the update processes for LLMs resemble those of conventional software. They were frustrated because software companies developing LLMs claimed to provide software but are not acting like it; despite the sentiment that LLMs are novel—too big, too complex, etc.—the “move fast and break things” mentality shouldn’t justify companies such as OpenAI’s claims that LLMs should be exempt from traditional practices.

Some participants noted that LLMs are large and open-domain, challenging developers to anticipate all use cases or assure specific outcomes, seeing that full transparency with LLMs isn’t feasible, as releasing a model doesn’t provide the same transparency and explainability as publishing source code in traditional software [9, 56, 72]. Given this real difference, one might then assume

that traditional software and APIs are beyond downstream uncertainties. Yet, benchmarking traditional software is not straightforward either [57]. Despite established practices like unit testing and automated oversight, unpredictable edge cases in software persist and verification remains a challenge [69, 70]; similarly, API revisions often break client software, with API refactorings causing 80% of disruptive code changes [7, 15, 54]. Thus, our work challenges readers to consider that LLMs, despite differences in their technical implementation and their size, have significant overlap with traditional software in how they are used, and thus the claim that LLMs justify departures from traditional software practices is largely unfounded.

5.2 What should proprietary LLM providers do?

Our participants were dissatisfied with the lack of transparency around updates, especially those from OpenAI. Their concerns included inadequate documentation from the developers, a lack of time to prepare for updates, insufficient advance notice regarding silent updates and deprecations, and a lack of migration guidance. These transparency issues diverged from traditional best practices, as highlighted by Vaniea [64]. Such problematic practices led participants to feel a sense of loss of control, thereby eroding their trust in the LLM provider.

To address these issues, we believe that proprietary LLM companies must take steps to fulfill their promises and start to act as software providers. Their practices must align with their responsibilities and users’ expectations. They ought not to deviate too far from traditional practices, and if they must, such digression should only occur in specific situations where genuine differences exist, such as the impossibility of full transparency. In summary, LLM companies should:

- (1) **Refrain from excusing bad practices with appeals to the novelty of AI.** LLM companies should not use the excuse of novelty to depart from standard software updating norms and user expectations. Our study suggests that users see past this rhetoric and that it erodes their trust in the provider.
- (2) **Provide advance warning of updates and establish norms.** Updates should be documented, and users should be able to see and revert to previous model versions if needed. Silent updates must be avoided to maintain user trust. Learning from software companies, LLM firms could adopt regular update schedules like monthly “Patch Tuesday” [63] to release updates.
- (3) **Deprecate gradually and accountably.** LLM companies need to follow a responsible model deprecation cycle, similar to the “deprecated-replace-remove” process used in software development. First, mark models as deprecated, then provide clear guidance for transitioning, and finally remove them. The timeline for this process in LLMs should match the typical 15-18 month duration observed in other software update norms [34, 39].
- (4) **Document detailed API usage and changes.** Companies should maintain up-to-date, accessible documentation for

API endpoints, parameters, return values, and usage examples, including change-logs for each release. Deprecated models or endpoints should be clearly labeled with a deprecation timeline.

- (5) **Document how and why models are updated.** This step includes tracking model development and dataset construction processes, and reporting potential limitations of the model and the dataset. These are practices advocated both by prior literature in machine learning [20, 42] and our participants to better predict how models might perform after updates.
- (6) **Onboard users.** Companies should guide users on adapting to updates and new features, including migration between models and best practices for smooth transitions. This involves offering clear tutorials, interactive resources, and community forums for support and knowledge sharing.
- (7) **Collaborate with users on benchmarking and transparency efforts.** Companies should develop tools and benchmarks to evaluate task-specific LLM behaviors, addressing the challenge of their opaque nature. This requires collaboration between companies and users, with companies gathering unique user cases to broaden benchmarking scope².

Overall, these actions would reduce user uncertainty and increase perceived control, leading to greater trust between LLM companies and users. Note that some of these are not very much different from traditional software, and that is the point. We felt the need to make these guidelines very clear for any readers who are developers of proprietary LLMs, who may, as we noted above, perceive their company or their work as outside of traditional software practice.

5.3 Generalizability to Other Proprietary Model Providers

OpenAI, with its ambitious expansion, the success of ChatGPT [17, 38] and significant investments such as Microsoft's \$10 billion infusion in 2023[19], solidifies its influential position as a proprietary model provider, driving trends and shaping industry standards. Consequently, the manner in which OpenAI addresses uncertainties in model updates and documentation not only influences user expectations but also guides the practices of other AI companies. The key role that OpenAI plays in being proprietary model providers shows that the need for improved transparency and adherence to conventional software update practices is applicable to other AI model providers. As OpenAI shapes the norms, other entities developing LLMs are likely to mirror these practices. For example, as mentioned, Anthropic's removed specific model names from their API after releasing Claude-2, rendering previous models inaccessible. If other LLM providers like Anthropic could adopt similar update practices to those causing frustrations among OpenAI users - such as abrupt model changes without clear communication, insufficient documentation, and a lack of transparency - there's a risk they could elicit the same type of user discontent.

²OpenAI, to their credit, appears to be attempting this [49].

5.4 LLMs as Infrastructure

A broader point, beyond the software update literature, is that participants seemed to perceive LLM companies as infrastructure providers. Participants' language evokes infrastructure; for instance, consider P11-a's worry that their system is reliant upon a service that "*can be cut off at any time*"—they could just have easily been talking about an unreliable water company turning off the water supply. As Leigh-Star notes [60, 61], infrastructure only appears when it breaks down. Another way to frame our findings is then that users, especially developers, expect providers to follow the norms of infrastructure companies. But because frequent updates cause frequent breakdowns (whether predictable or unpredictable), LLM companies are constantly disrupting the notion that (their) LLMs are infrastructure, leading some users to consider their services as "prototyping" tools instead. They then look to alternatives which actually live up to the promise of infrastructure, like local embedded models, which are reliable and consistent. The rhetoric from companies and research reinforces the notion that LLMs are infrastructure; for instance, consider the fact that LLMs are doubly called "foundation models" [23]. A "foundation" in common parlance is "a body or ground upon which something is built up or overlaid" [40], implying stability. A foundation that is constantly shaking is not a foundation.³

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A APPENDIX: INTERVIEW GUIDE

A.1 Introduction

- (1) Welcome and thank the participants for their time.
- (2) Briefly explain the purpose of the study.
- (3) Assure the participant of their anonymity and the confidentiality of their responses.
- (4) Obtain consent to proceed with the interview.

A.2 Background Information

- (1) Ask the participant about their experience with LLM/LLM APIs, including:
 - (a) Which applications they have used (e.g., ChatGPT, GPT-4, Claude).
 - (b) What is your motivation for using such applications?
 - (c) How long they have been using these applications.
 - (d) The frequency of their usage.

A.3 Perceived Impact of Generative AI Updates

- (1) Ask the participant to share their experiences with updates to LLMs, focusing on: [For each X model they mentioned above, ask:]
 - (a) Can you recall a time the [X applications]' performance or functionality changed, or that you perceived that it changed? [If yes]: Describe that to me. How did you become aware of the update(s) to X? Think of how developers deployed updates to X. What could developers have done differently? [Up to 3 times] Can you think of another example?
 - (b) How do they adapt to the new infrastructure? What did you do differently when the models are updated?
 - (c) If any, can you recall a similar experience of using traditional apps that are not based on Generative AIs?
- (2) Can you provide an example of an LLM update that had a significant impact on your user experience? Please describe the changes you noticed.

A.4 Ethical and Responsible Updates

- (1) Discuss the participant's thoughts on what constitutes ethical and responsible LLM updates, including:
 - (a) In your opinion, When faced with updates, what features of the application helped you improve your user experience and what did not?
 - (b) What other features that are currently not available that you imagine could help your experience?
 - (c) In your own words, what do you think constitutes ethical and responsible Generative AI model updates?
- (2) Hypothetical scenario: What would you think will happen if the model that you are using is going to be deprecated?

A.5 User Control and Transparency

- (1) Ask the participant about their preferences regarding control and transparency in Generative AI-based application updates:
 - (a) How important is it for you to have control over the updates in Generative AI-based applications? Why?

- (b) How would you feel if updates to the AI model happened without your knowledge or consent? Can you provide an example of how this might affect your user experience?
- (c) What level of transparency do you expect from developers when they update LLMs? How might this impact your trust in the application?
- (d) In your opinion, should developers provide a way for users to customize the AI's behavior according to their preferences? Why or why not?

A.6 Conclusion

- (1) Thank the participant for their valuable input.
- (2) Invite the participant to share any final thoughts or suggestions on the topic.
- (3) Reiterate the confidentiality of their responses.
- (4) Inform the participant about any follow-up communication, if applicable.

B APPENDIX: CODEBOOK

Table 1: Codebook used in the open coding process

Theme	Sub-Theme	Description
Uncertainties in Model Updates	Perceived Model Degradation	Instances where users note a decline in performance without prior announcement.
	Model Optimization with Side Effects	Updates aimed at improving certain aspects inadvertently changed model performance or behaviors.
	Abrupt Model Deprecation	Model deprecation that is unannounced or with little notice.
	Insufficient Documentation	Inadequate guidance or documentation accompanying updates or deprecation.
Consequences of Uncertainties	Personalized Solution Development	The need for users to devise their own fixes due to the lack of clear guidelines.
	Lost Trust in Proprietary LLM APIs	The impact of uncertainties on user trust, particularly around transparency, reliability, and viability.
	Exploration of Alternatives	Consideration or movement towards open-source or locally deployed models due to frustrations with proprietary LLMs.
Participant Recommendations	Transparency Enhancements	Calls for more detailed benchmarks, transparency in training data, and clear documentation of update changes.
	Adoption of Conventional Software Practices	Recommendations for LLM companies to adopt more traditional software company standards.
	Addressing Novelty Excuses	Critiques of LLM companies justifying shortcomings by emphasizing AI's novelty.
	Addressing Difference between LLM and Traditional Softwares	Discussions that touch upon the technical difference between LLMs and traditional software.